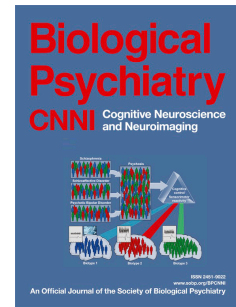


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Effects of maternal psychopathology and education level on neurocognitive development in
infants of adolescent mothers living in poverty in Brazil

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Abstract

Background

Adolescent motherhood remains common in developing countries and is associated with risk-factors that adversely impact infant neurodevelopment, including poverty, low maternal education and increased maternal psychopathology. Yet, no published work has assessed how these factors affect early brain development in developing countries.

Methods

This pilot study examined effects of maternal psychopathology and education on early neurocognitive development in a sample of adolescent mothers ($n = 50$, final $n = 31$) and their infants living in poverty in São Paulo, Brazil. Maternal symptoms of anxiety, depression and attention-deficit/hyperactivity disorder (ADHD) and education level were assessed during pregnancy. Infant neurocognitive development was assessed at age 6 months with oscillatory power and functional connectivity in the theta (4-6Hz), alpha (6-9Hz) and gamma (30-50Hz) frequencies derived from resting-state electro-encephalography (EEG), temperament (negative affect, attention/regulation), and cognitive, language and motor skills. Cluster-based permutation testing and graph-theoretical methods were used to identify alterations in oscillatory power and connectivity that were associated with maternal psychopathology and education. Correlations between power and connectivity alterations were examined in relation to infants' overt cognitive-behavioural abilities.

Results

Increased maternal anxiety and lower maternal education were associated with weaker oscillatory connectivity in alpha-range networks. Infants with the weakest connectivity in the alpha network associated with maternal anxiety also showed the lowest cognitive ability. Greater maternal anxiety and ADHD were associated with increased absolute and relative theta power.

Conclusions

Our findings highlight the importance of addressing maternal psychopathology and improving education in poor adolescent mothers to prevent negative effects on infant neurodevelopment.

Background

Adolescent motherhood remains common in developing countries and is associated with increased prevalence of risk-factors that are known to adversely impact infant neurodevelopment. In Brazil for example, the adolescent birth rate is 60.5/1000 compared to 14.4/1000 and 22.3/1000 in the UK and US respectively [1]. These young mothers frequently live in impoverished environments and show high rates of mental health problems, including depression, anxiety, impulsivity and suicidal behaviour, which are often undetected and untreated by healthcare professionals [2-3]. These girls are often disconnected from school and find it increasingly difficult to continue their scholarship during pregnancy, resulting in low education levels [4-5]. Increased rates of neurodevelopmental disorders such as attention-deficit/hyperactivity disorder (ADHD) are found in adolescent mothers in developed countries [6], though this association has not been studied in developing countries. Outside of the context of adolescent motherhood, these factors have been robustly associated with poorer cognitive, language and socio-emotional development and increased rates of internalising and externalising disorders in offspring [7-8].

Given the high rates of these risk-factors amongst adolescent mothers in low-resource countries, infants born to these young mothers represent a high-risk population for impaired neurocognitive development. Yet, few studies have investigated the development of these infants. In Brazil, the handful of studies have reported poorer motor development in the first 18 months of life [9] and increased internalising and externalising behaviour problems in toddlerhood [10] in infants of adolescent mothers. It is unclear, however, which specific risk-factors contribute to these early developmental problems. Furthermore, the mechanisms by which these risks result in impaired infant development have not been investigated. One explanation for the latter question is that maternal psychopathology coupled with poverty and poor education alters early neurodevelopmental processes, particularly the formation of brain

networks which provide the structure for acquiring motor, cognitive and socio-emotional skills [11]. In turn, early disruptions to brain development lead to atypical developmental trajectories [12]. Consistent with this account, atypicalities in brain structure, function, and functional connectivity in the first year of life have been associated with low socio-economic background, maternal psychopathology and low parental education in infants of adult mothers [13-16].

To our knowledge, no published work has examined the effects of maternal risk-factors on neurodevelopment in infants of adolescent mothers in developing countries. Understanding precisely how brain function is affected in these infants and which maternal risk-factors brain alterations relate to, e.g. psychopathology or low education, will be crucial in designing targeted, effective interventions to prevent impaired development in these infants. This knowledge will also be important for supporting public policies designed to reduce social disadvantages from the beginning of life.

In this pilot study we took the first steps to investigating these issues by examining how maternal psychopathology (symptoms of depression, anxiety and ADHD) and education level affect early brain development in infants of adolescent mothers living in impoverished urban regions of São Paulo, Brazil. Infant brain function was assessed using resting-state electro-encephalography (EEG) from which we derived measures of oscillatory power and phase synchronisation (functional connectivity). In EEG, *oscillatory power* indexes the magnitude of activity across populations of neurons, recorded at the scalp. The frequency characteristics of oscillatory power have been robustly linked with different neurocognitive processes in infants, older children and adults [17-18]. For example, power in the mid-range alpha frequency (primarily 6-9Hz in infants; 8-12Hz in older individuals) is associated with cortical inhibition and attention [19-22], low-frequency theta power (4-6Hz in infants; 4-8Hz in older individuals) is implicated in attentional and regulatory processes [22-25], and high-

frequency gamma power (>30Hz) is associated with integrative perceptual processes and cognitive and language development [26-28]. In infants, atypicalities in oscillatory power, including reduced alpha and gamma power and increased theta power, have been found in relation to maternal psychopathology, socio-economic and psychosocial deprivation and later-emerging developmental problems [13-14,29]. Another important characteristic of oscillatory activity is *phase synchronisation*, which reflects the extent to which signals from different neural populations oscillate in synchrony (with the same timing). Phase synchrony is believed to be a key neuronal communication mechanism which coordinates activity across different brain regions (functional connectivity), facilitating the formation of functional neural networks [30]. While the specific brain regions involved in such networks cannot be localised with EEG due to the technique's low spatial resolution, the frequency at which phase synchronisation occurs is, like oscillatory power, associated with particular neurocognitive functions [17,31]. Further, disruptions to oscillatory networks, such as reduced phase synchrony in the alpha and gamma frequencies and atypically increased phase synchrony in the theta frequency, have been found to index exposure to adverse environments, maternal psychopathology and early developmental problems in infancy [32-34]. Oscillatory power and phase synchronisation are therefore valuable methods for investigating the influence of maternal variables on neural activity and the temporal dynamics of neural networks in early neurodevelopment. Following previous work [13-14,21-23,27-29,31-34], we focused on oscillatory power and phase synchronisation (hereafter, '*connectivity*') in the infant theta (4-6Hz), alpha (6-9Hz) and gamma (30-50Hz) frequencies.

We used cluster-based permutation testing and graph theoretical methods to test the following hypotheses: 1) higher maternal psychopathology and lower maternal education would be associated with lower oscillatory power and connectivity in the alpha and gamma frequencies and higher power and connectivity in the theta frequency in infants; 2) these

alpha/gamma reductions and theta increases would in turn be associated with poorer cognitive, language, emotional, motor and attentional/regulatory abilities in infants. Given the dearth of information on neurodevelopment in infants of adolescent mothers in developing countries, we also explored oscillatory activity patterns directly associated with infants' developmental abilities. Based on previous research with infants of adult mothers, we expected stronger power and connectivity in the alpha and gamma frequencies to be associated with better cognitive, language and motor abilities; for the theta frequency, we expected stronger power and connectivity (at different scalp regions to those associated with maternal psychopathology and education) to be associated with better attentional/regulatory abilities [21-22,27-28].

Methods

Participants

Fifty adolescent mothers and their infants participated in this study. Adolescents were recruited during their pregnancies from primary healthcare units in poor western regions of São Paulo, Brazil. These regions are characterised by high rates of urban violence, widespread slums and adverse living conditions (lack of sanitation, poverty). At recruitment, all adolescents were aged 14-19 years, pregnant for the first time, between 8-16 weeks gestation, living in western regions of São Paulo, and of low socio-economic status (classes C-D/E according to the Criteria for Economic Classification assessed by the Brazilian Association of Research Companies questionnaire [35]). The adolescents were taking part in a larger randomised controlled trial (NCT0280718) of a home visitation intervention designed to improve parenting skills and infant development [36]. Twenty-five adolescents received the intervention and 25 care-as-usual. The current dimensional analyses were conducted on the Intervention and Control groups combined with intervention group covaried

in analyses (see Supplementary Materials for further details concerning effects of the intervention).

Infants completed EEG and cognitive-behavioural assessments at age 6 months. Two infants were born one month prematurely (retained in analysis); the remaining 48 infants were full-term. All infants were without neurological or genetic problems. Usable EEG data were not acquired from all 50 infants (detailed below) and final analyses were conducted on a sample of $n = 31$. The infants who did and did not provide usable EEG data did not differ significantly in infant or maternal characteristics (see Supplementary Materials for retention analysis). Characteristics of the full and analysis samples are shown in Table 1. Written informed consent was obtained from all adolescent mothers and, if aged <18 years, their parent/guardian. The study was approved by the ethical review boards of the University of São Paulo Medical School (ref: 052/15) and the São Paulo Municipal Health Department.

[Table 1]

Measures

Maternal psychopathology and education

Maternal psychopathology and education level were assessed at recruitment into the study. Education level was classified into five categories (see Table 1). Symptoms of anxiety and depression were assessed using the Brazilian Portuguese versions of the Beck Anxiety and Depression Inventories (BAI/BDI) [37], which are 21-item rating-scales that assess the presence and severity of anxiety and depression symptoms. Scores range from 0-63; higher scores reflect more severe symptoms. Maternal ADHD symptoms were assessed with the Brazilian Portuguese version of the Adult ADHD Rating Scale (ASRS) [38], which contains 18 questions measuring the frequency of recent ADHD symptoms and is based on

DSM-IV-TR [39] diagnostic criteria for adult ADHD. Scores range from 0-72; higher scores indicate more severe ADHD. The BAI, BDI and ASRS were administered to adolescents in interview format by trained psychologists. The total score for each measure was used in analysis.

Cognitive-behavioural measures of infant development at 6 months

The Bayley Scales of Infant Development-Third Edition (BSID-III) [40], a standardised instrument for infants aged 1-42 months, was administered to all infants by trained psychologists. Age-normed composite scores (mean 100, SD 15) for cognitive, language and motor ability were used in analysis. Infant temperament was assessed with the Brazilian Portuguese version of the Infant Behavior Questionnaire – Revised (IBQ-R) [41] administered to mothers in interview format by psychologists. The IBQ-R is a 191-item scale in which parents rate how often (never-to-always) their infant exhibits particular behaviours in everyday contexts. Items are averaged to yield subscales and factors indexing different temperament dimensions. In this study, the factors *negative affect* and *orienting/regulation* were used. Higher negative affect scores reflect greater negative emotions (fear, sadness, distress). Higher orienting/regulation scores reflect better attentional/regulatory abilities (longer attending/orienting to objects, higher soothability).

Resting-state EEG at 6 months

EEG recording and pre-processing

Infants completed a 3-minute resting-state EEG recording while sitting on their mother's lap approximately 65cm in front of a computer screen in a dimly lit room. A video of abstract shapes was shown throughout recording to engage infants' attention. EEG data were recorded using a 128-channel Geodesic Sensor Net and a NetAmp 200 DC-coupled

amplifier [42]. The data were referenced online to electrode Cz, sampled at 500Hz and bandpass filtered between 0.1-100Hz. EEG data were pre-processed offline using Brain Vision Analyzer v2.1 [43]. Electrodes around the rim of the net were contaminated by excessive artefact and removed from all participants, leaving 80 electrodes in analysis. The data were filtered using 0.1Hz high-pass, 60Hz low-pass 24dB/Oct Butterworth filters with a 60Hz notch filter for residual electrical line noise. Periods of data with excessive noise or during which the infant was crying or reacting to external stimuli were excluded. Remaining flat or noisy channels were removed and interpolated using spherical spline interpolation prior to re-referencing to the average reference. Independent Components Analysis was used to identify and remove ocular artefact components after which the data were segmented into 2-second non-overlapping epochs. Epochs with remaining artefacts (amplitudes $\pm 150\mu\text{V}$) were excluded. Infants with fewer than 20 artefact-free epochs were excluded from analysis. Lastly, a Laplacian transform using spherical splines [44] ($\lambda: 1\text{e-}05$; order of splines: 4; degree of legendre polynomials: 20) was applied to control for volume conduction.

EEG data from fourteen infants were unusable due to technical problems with the EEG system resulting in corrupted data files ($n = 4$) or poor-quality recordings ($n = 10$). A further five infants had fewer than 20 artefact-free epochs for analysis and were also excluded, leaving a final sample for analysis of $n = 31$ mother-infant dyads (17 Intervention, 14 Control).

Computation of oscillatory power and connectivity

The cleaned, Laplacian-transformed epochs were exported to FieldTrip [45] within the MATLAB [46] environment for computation and analysis of oscillatory power and connectivity. In FieldTrip, the 2-second epochs were subjected to Fast Fourier Transform with a 10% Hanning window taper to obtain power estimates and fourier coefficients for the

1-50Hz range at 1Hz intervals. Power estimates were averaged across frequency steps in the theta (4-6Hz), alpha (6-9Hz) and gamma (30-50Hz) ranges (see Supplementary Materials for further information on frequency bands), resulting in power-spectra containing the average absolute theta, alpha and gamma power at each electrode. Relative power spectra were also computed (absolute theta/alpha/gamma band power divided by absolute 1-50Hz broadband power). Absolute and relative power-spectra (six per participant) were used in statistical analyses.

Oscillatory connectivity (phase synchronisation) was quantified by the debiased weighted phase lag index (dwPLI, see [47] and Supplementary Materials), computed from fourier coefficients at each 1-50Hz frequency step between each pair of electrodes across epochs. This resulted in one 80x80 adjacency matrix per frequency step per participant, where matrix element (ij) holds the phase synchronisation (dwPLI) of signals between i and j electrodes. Adjacency matrices were averaged across frequency steps to obtain one 80x80 adjacency matrix for connectivity in the theta, alpha and gamma frequency bands. The resulting matrices (three per participant) were used in statistical analysis.

Statistical analysis

The hypothesis that maternal psychopathology and education would be associated with decreases (alpha/gamma) and increases (theta) in infants' oscillatory activity (hypothesis 1) was tested by examining associations between maternal variables (anxiety, depression, ADHD, education level) and infants' oscillatory power (relative and absolute theta, alpha and gamma power) and connectivity (dwPLI in theta, alpha and gamma bands). For absolute and relative power, cluster-based permutation testing [48] was conducted in FieldTrip to identify clusters of electrodes at which power was significantly ($p < 0.05$) positively or negatively associated with maternal psychopathology and education while controlling for multiple

comparisons (80 power values per power-spectra). For oscillatory connectivity, Network Based Statistic (NBS) [49] was used to identify oscillatory neural networks, defined according to graph theory [50] as topologically connected clusters of nodes (electrodes) based on the strength of their edges (oscillatory synchrony between electrodes, dwPLI), that were significantly ($p < 0.05$) associated with maternal psychopathology and education while controlling for multiple comparisons (6,400 dwPLI values per adjacency matrix). Further details of the cluster-based permutation and NBS methods are provided in the Supplementary Materials. Additionally, whole-brain connectivity, defined as the average dwPLI across all electrodes, was computed for each frequency band (theta, alpha, gamma) and used as an outcome variable in regression models, with maternal variables predicting whole-brain connectivity (all variables were mean-centred prior to regression analysis). All analyses were conducted while covarying Group (Intervention, Control) to control for possible intervention effects. Subsequently, analyses were repeated covarying maternal age (years) to control for biological maturity of the mother, infant age (weeks, corrected for prematurity) to control for age-related changes in neural activity, and SES to control for variability in socio-economic deprivation. We also modelled interactions with Group (see Supplementary Materials for details).

The hypothesis that greater decreases (alpha/gamma) and increases (theta) in oscillatory power/connectivity would be associated with poorer cognitive-behavioural development (hypothesis 2) was tested by first computing the average power in clusters and average connectivity in networks that were significantly associated with maternal variables. Next, partial Spearman correlations were computed between those power/connectivity metrics and infant cognition, language, motor, attention/regulatory ability and negative affect. Correlations were computed controlling for Group and repeated controlling for maternal/infant age and SES.

Our exploratory analysis examining oscillatory activity directly associated with infants' cognitive-behavioural development was conducted using cluster-based permutation testing, NBS and regression (as described above for hypothesis 1) to identify patterns of power and connectivity that were related to infants' cognition, language, motor and attention/regulatory ability and level of negative affect.

Results

Infant oscillatory power and maternal psychopathology and education

Cluster-based permutation testing revealed significant positive associations between absolute theta power at frontal/posterior electrode clusters and maternal anxiety ($p=0.04$) and absolute theta power at posterior clusters and maternal ADHD symptoms ($p=0.03$) while controlling for Group (Table 2, Figure 1). Similar positive associations were revealed between relative theta power at fronto-central clusters and maternal anxiety ($p=0.009$) and relative theta power at frontal/posterior clusters and maternal ADHD symptoms ($p=0.008$) while controlling for Group (Table 2, Figure 2). These effects remained significant while controlling for maternal and infant age and SES ($p<0.05$). There were no further significant associations and no interactions with Group ($p>0.06$).

[Table 2, Figures 1&2]

Infant oscillatory connectivity and maternal psychopathology and education

NBS revealed networks in the alpha range in which connectivity was significantly associated with maternal anxiety and education level while controlling for Group; weaker connectivity in these networks was associated with greater maternal anxiety ($p=0.03$) and lower maternal education ($p=0.01$) (Table 2, Figure 3). The network associated with anxiety

remained significant controlling for maternal age ($p=0.04$), marginal covarying SES ($p=0.055$) but non-significant covarying infant age ($p=0.13$). The network associated with maternal education remained significant when covarying maternal and infant age and SES ($p<0.02$). There were no further significant associations and no interactions with Group ($p>0.06$). Whole-brain connectivity was not associated with maternal variables ($F<2.49$, $p>0.10$).

[Figure 3]

Infant EEG and cognitive-behavioural abilities

Partial Spearman correlations (controlling for Group) were computed between power and connectivity measures that were significantly associated with maternal psychopathology and education (average absolute and relative theta power in clusters associated with maternal anxiety and ADHD, average connectivity in alpha-range networks associated with maternal anxiety and education level) and infants' cognitive, language, motor, attentional/regulatory abilities and negative affect. There was a significant association between infant cognitive ability and connectivity in the alpha-range network that was related to maternal anxiety ($\rho(28)=.377$, $p=0.04$); infants with weaker connectivity in this network had lower cognitive skills (Figure 4a). This association remained significant when controlling for maternal/infant age and SES ($\rho>.462$, $p<0.01$). There were no further significant associations ($\rho<.334$, $p>0.07$).

[Figure 4]

We next examined how oscillatory power and connectivity were associated with infants' cognitive-behavioural abilities directly. Cluster-based permutation testing showed that increased negative affect was associated with greater relative alpha power at central clusters ($p=0.04$ covarying Group, Table 2, Figure 4b; $p<0.04$ covarying infant/maternal age and SES). NBS revealed no significant networks associated with cognitive-behavioural abilities (all $p>0.06$). However, whole-brain connectivity in the theta range was positively associated with infants' attentional/regulatory ability ($F(2,28)=3.42$, $p=0.047$, $AdjR^2=.15$ covarying Group), with stronger theta connectivity across all electrodes associated with better attention/regulation ($\beta=.44$, $t=2.60$, $p=0.015$, Figure 4c); this association remained significant when covarying infant and maternal age ($F>2.93$, $p<0.05$) and at trend-level when covarying SES ($F=2.60$, $p=.07$). There were no further significant associations and no interactions with Group ($p>0.08$).

Discussion

This pilot study examined effects of maternal psychopathology and education on neurodevelopment in infants of poor adolescent mothers living in poverty in Brazil. Increased maternal anxiety and lower maternal education were associated with weaker connectivity in oscillatory networks in the alpha frequency. Since connectivity in functional brain networks increases over the first year of life in typical development [31,52], our findings of reduced connectivity in association with increased maternal anxiety and lower maternal education suggest that these maternal factors disrupt this process, perhaps hindering the formation of functional neural networks. Importantly, infants with the weakest connectivity in the network associated with maternal anxiety also showed the lowest cognitive ability. This finding suggests that the disrupted connectivity related to maternal anxiety had a negative impact on infants' overt neurocognitive ability, although the direction of causality cannot be determined from the current analyses. These connectivity findings are

in line with previous work reporting altered alpha activity in infants of anxious mothers [14] and reduced structural brain development in infants of parents with lower education [16]. Our study is the first, however, to report that these maternal factors are associated with reduced functional connectivity in oscillatory brain networks in infants of poor adolescent mothers. That these disruptions were restricted to the alpha frequency is informative about the function of the under-connected networks. Alpha oscillations play a key role in cortical inhibition and are likely mediated by inhibitory GABAergic signalling [19]. Evidence indicates that inhibitory neuronal signalling is fundamental for cortical reorganisation and guiding functional neural network formation early in neurodevelopment [53]. The weaker alpha connectivity associated with maternal anxiety and low education may therefore reflect disruptions to inhibitory signalling mechanisms and the architectural roles they play in brain circuit development.

We also found that higher levels of maternal anxiety and ADHD symptoms were associated with increases in absolute and relative theta power in infants. These findings are consistent with previous reports of increased theta power in infants of adult anxious mothers [54] and, though no published work has examined effects of maternal ADHD on infant brain function, with reports of elevated theta power indexing genetic risk for ADHD in adolescents [55]. Given the role of theta oscillations in attentional and regulatory processes [22-25] and the involvement of these cognitive functions in anxiety and ADHD [56-58], the current findings may reflect a vulnerability to emotion/attention regulation problems that is shared between maternal risk for anxiety and ADHD. However, these patterns of theta power were not correlated with infants' overt attentional/regulatory ability or levels of negative emotionality. Still, perhaps at this early point in infancy alterations in oscillatory power associated with maternal ADHD and anxiety do not yet translate to alterations in overt

behaviour. It will be important to investigate how these early oscillatory atypicalities relate to longer-term developmental abilities.

Our exploratory analysis revealed patterns of oscillatory power and connectivity associated with infants' developing cognitive-behavioural abilities that are consistent with findings in other populations of infants. Infants with higher attentional/regulatory abilities showed stronger whole-brain connectivity in the theta range, in line with findings in infants of adult mothers reporting increased theta oscillations during attentional control [22-23]. Infants with higher levels of negative affect showed increased relative alpha power, consistent with previous findings in infants of adult mothers [59]. Our findings add to the infant development field by showing that oscillatory dynamics associated with early neurocognitive abilities are replicable in previously unstudied samples of infants from developing countries.

Clinical implications

Our findings provide the first empirical (albeit suggestive) evidence from developing countries indicating that pregnant adolescents' mental health problems are associated with altered development of their infants' oscillatory neural activity in the first 6 months of life. A recent epidemiological study in Southern Brazil reported that 22.5% of pregnant adolescents met diagnostic criteria for at least one mental disorder and that symptoms were undetected (and untreated) in 80% of cases [2]. The current findings highlight the importance of addressing this situation so that these young mothers receive appropriate mental health interventions, not only to improve their own health and well-being but also to prevent adverse effects on their infants' neurodevelopment. In addition to targeting maternal mental health problems, our findings indicate that interventions for young mothers in developing countries should focus on improving education. Clearly, depending on their age adolescent mothers

will not have had the opportunity to complete the higher levels of their education. Still, evidence suggests that Brazilian pregnant adolescents have poorer educational attainment than their age-matched peers [4-5], and in our sample the mean age of the mothers who had only completed elementary school was only slightly younger than those who had completed middle school (15.8 vs. 16.5 years). Our findings could also be used to support public policies designed to reduce inequality and the challenges adolescent mothers in developing countries face, including poverty and lack of social support, which may in turn reduce the likelihood that these girls experience mental health problems and improve their infants' development.

Limitations

Our modest sample size and high data loss precluded more complex statistical modelling. Consequently, we could not examine interactive relationships between maternal factors and infant neurodevelopment. Future work in larger samples is needed to replicate our findings and model different pathways between maternal and infant variables. Relatedly, our analysis of interactions with intervention group were likely under-powered. We did not examine how factors such as maternal sensitivity affected associations between maternal risk-factors and infant neurodevelopment. This is important since previous work indicates maternal sensitivity can mediate associations between risk-factors and infant development in adolescent mothers [60]. Our infant data were from one time-point only and future longitudinal work is needed to examine whether early disruptions to infant brain function associated with maternal risk-factors lead to altered developmental trajectories.

Conclusions

This pilot study provides the first evidence from developing countries that maternal psychopathology and low maternal education are associated with alterations in oscillatory neural activity in infants of adolescent mothers. These findings could be used to tailor appropriate interventions and to support public policies aimed at alleviating social disadvantages in vulnerable groups.

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Table 1 Sample characteristics. Means (SD) are presented.

	Recruited sample (<i>n</i> = 50)	Analysis sample (<i>n</i> = 31)
<i>n</i> Intervention, <i>n</i> Control	25, 25	17, 14
<i>Maternal measures at baseline</i>		
Age (years)	16.86 (1.37)	16.61 (1.54)
SES (<i>n</i> C, <i>n</i> DE) ^a	32, 18	19, 12
Anxiety (BAI)	10.68 (7.56)	12.29 (8.02)
Depression (BDI-II)	13.22 (7.73)	14.45 (8.18)
ADHD (ASRS)	24.30 (10.85)	25.74 (11.74)
Maternal education level (<i>n</i> L1, <i>n</i> L2, <i>n</i> L3, <i>n</i> L4, <i>n</i> L5) ^b	0, 7, 38, 5, 0	0, 6, 22, 3, 0
<i>Infant measures at 6 months</i>		
Sex (<i>n</i> girls, <i>n</i> boys)	22, 28	15, 16
Age (weeks)	26.76 (1.12)	26.74 (1.15)
Bayley Cognitive Composite	101.70 (11.19)	100.48 (10.52)
Bayley Language Composite	88.76 (12.55)	90.42 (7.75)
Bayley Motor Composite	101.02 (11.50)	100.10 (11.34)
IBQ-R Negative Affect	3.96 (0.87)	4.03 (0.85)
IBQ-R Orienting/Regulation	5.13 (0.67)	5.08 (0.67)
<i>n</i> EEG epochs	----	46.81 (18.93)

SES C/DE = socio-economic status classes C or DE. BAI = Beck Anxiety Inventory total score. BDI-II = Beck Depression Inventory II total score. ASRS = Adult ADHD Rating Scale total score. Maternal Education level L1 = illiterate/elementary school incomplete, L2 = elementary school complete/middle school incomplete, L3 = middle school complete/high school incomplete, L4 = high school complete/university incomplete, L5 = university complete. Bayley = Bayley Infant Scales of Development III. IBQ-R = Infant Behavior Questionnaire – Revised. *n* EEG epochs = number of resting-state EEG epochs included in analysis.

Table 2 Statistical test results for cluster-based permutation tests and network based statistic

Significant association (covarying Group)	Degrees of freedom (df)	Cluster statistic (effect size)	Network statistic (effect size)
<i>Maternal anxiety and infant absolute theta power</i>	30	44.83 (0.42)	-----
<i>Maternal ADHD and infant absolute theta power</i>	30	43.55 (0.46)	-----
<i>Maternal anxiety and infant relative theta power</i>	30	134.18 (0.46)	-----
<i>Maternal ADHD and infant relative theta power</i>	30	102.57 (0.51)	-----
<i>Maternal anxiety and infant alpha connectivity</i>	30	-----	3.47 (0.62)
<i>Maternal education and infant alpha connectivity</i>	30	-----	3.41 (0.61)
<i>Infant negative affect and relative alpha power</i>	30	17.75 (0.40)	-----

Cluster statistic (effect size) = the maximum cluster statistic (summed cluster statistics) that was subjected to permutation testing in cluster-based permutation tests and its effect size (Cohen's d estimate). *Network statistic (effect size)* = the average network statistic for the significant connections in networks identified by network based statistic and its effect size (Cohen's d estimate). See the Supplementary Materials for full details of the procedure for computing the cluster and network statistics and effect size estimates.

Figure titles and captions

Figure 1 Associations between infant absolute power and maternal anxiety and ADHD symptoms

Topographical plots (top and bottom left) show the clusters of electrodes (indicated by X signs) at which infants' absolute oscillatory power in the theta frequency was significantly positively associated with maternal anxiety (Panel A) and ADHD (Panel B) symptoms. Colour bars represent the statistical strength (T-value) of the association between oscillatory power at different scalp regions and maternal anxiety/ADHD, with higher T-values reflecting a stronger statistical association. Scatterplots (top and bottom right) display the positive associations between absolute theta power in the significant clusters and maternal anxiety (A) and ADHD (B). Note that absolute power values are large due to the Laplacian transform applied to the data prior to computation of spectral power (μV^2).

Figure 2 Associations between infant relative power and maternal anxiety and ADHD symptoms

Topographical plots (top and bottom left) show the clusters of electrodes (indicated by X signs) at which infants' relative oscillatory power in the theta frequency was significantly positively associated with maternal anxiety (Panel A) and ADHD (Panel B) symptoms. Colour bars represent the statistical strength (T-value) of the association between oscillatory power at different scalp regions and maternal anxiety/ADHD, with higher T-values reflecting a stronger statistical association. Scatterplots (top and bottom right) display the positive associations between absolute theta power in the significant clusters and maternal anxiety (A) and ADHD (B).

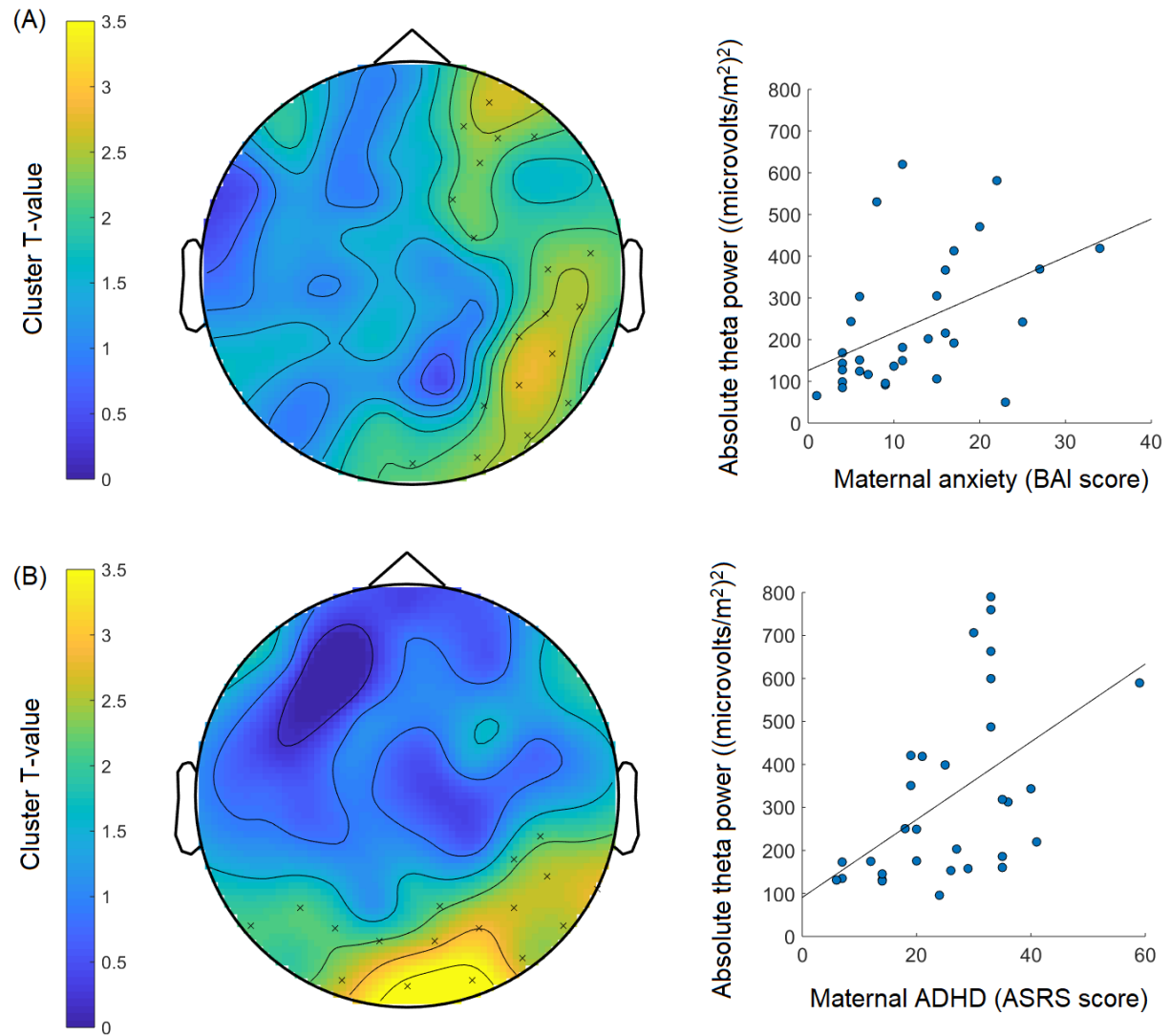
Figure 3 Associations between infant oscillatory network connectivity and maternal anxiety and education level

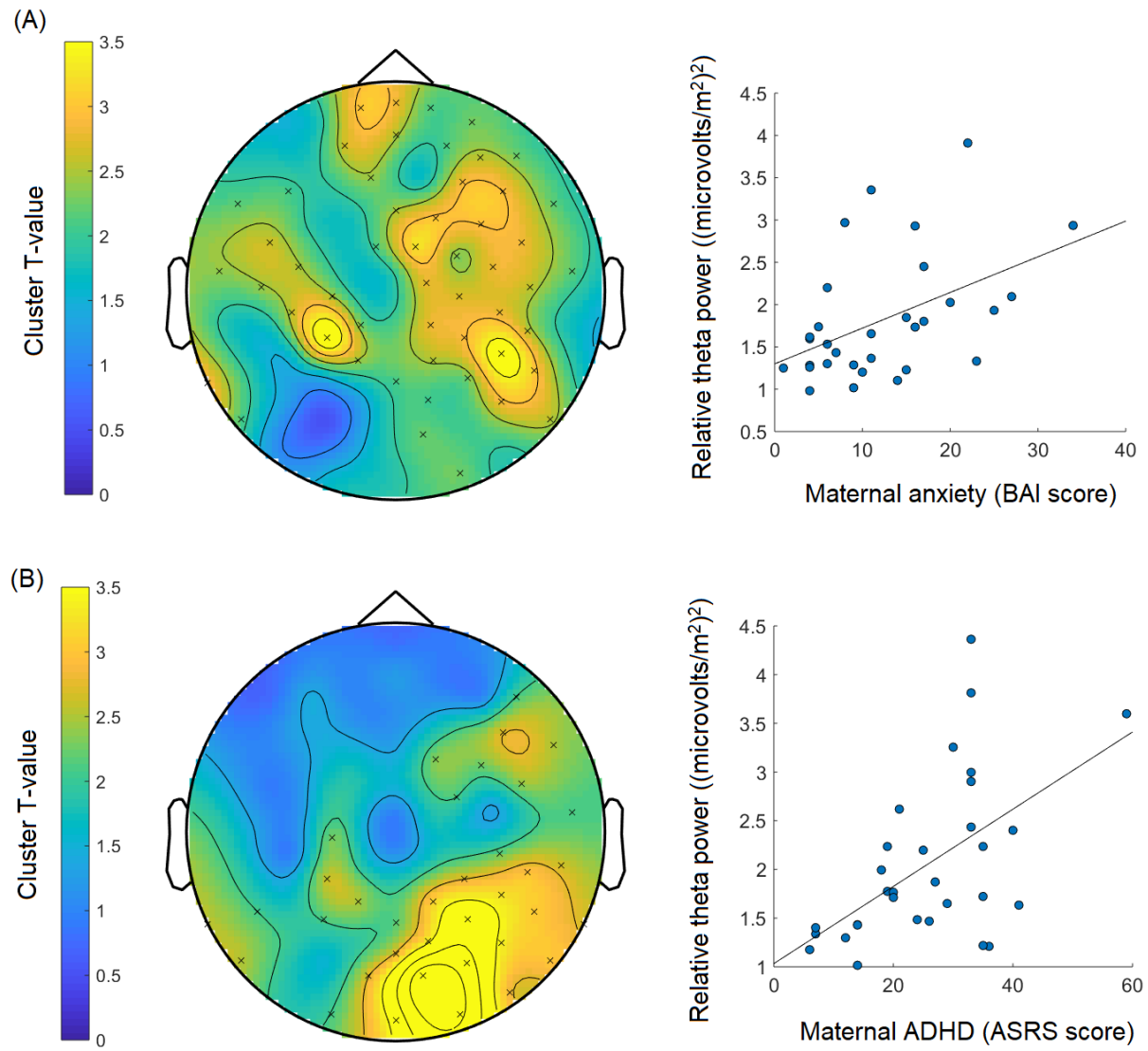
Panel (A) shows the oscillatory network (left) in the alpha frequency in which connectivity (dwPLI) was significantly negatively associated with maternal anxiety; black circles represent the nodes (electrodes) of the network and blue lines represent the edges (functional connections between nodes, indexed by oscillatory synchrony – dwPLI). All edges in the network showed reduced dwPLI (reduced connectivity) in association with greater maternal anxiety symptoms. The networks were visualised using BrainNet Viewer [51]. The scatterplot (right) displays the negative association between connectivity (average dwPLI in the network) and maternal anxiety symptoms. Note that the dwPLI can take small negative values due to the debiasing procedure. Panel (B) shows the oscillatory network in the alpha frequency (left) that was significantly associated with maternal education level. All connections in this network showed weaker connectivity (lower dwPLI values) in association with lower levels of maternal education, as illustrated in the scatterplot to the right.

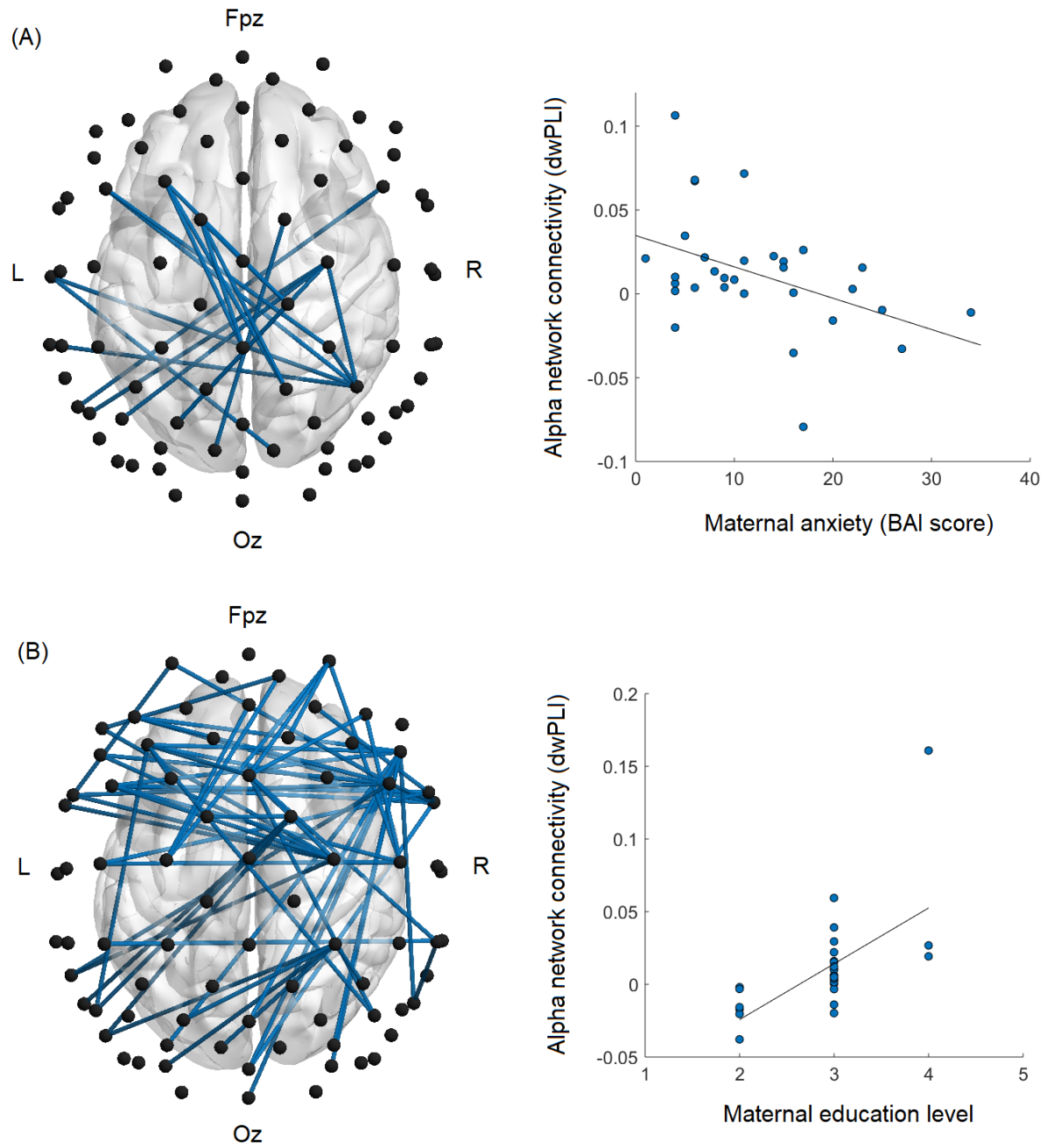
Figure 4 Associations between infant EEG and cognitive-behavioural abilities

Panel (A) shows the positive association between infants' relative alpha power and level of negative affect; the topographical plot displays the clusters of electrodes (indicated by X signs) at which this association was significant (colour bar indicates the statistical strength – T-value – of this association) and scatterplot displays the positive association between

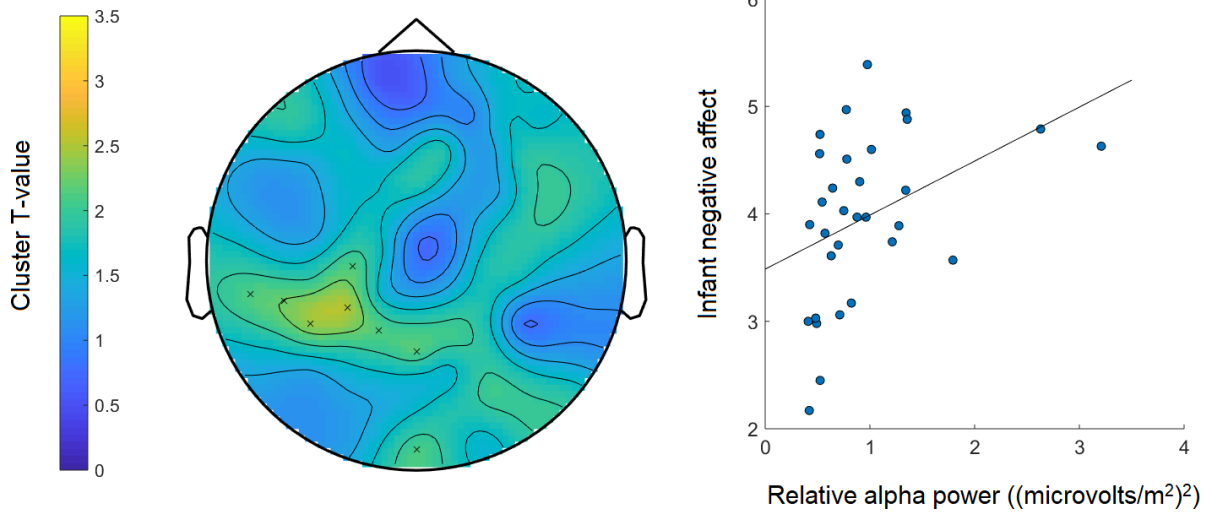
relative theta power in the significant clusters and infant negative affect scores. Panel (B) shows the significant association between infant cognition and connectivity in the alpha-range network that was associated with maternal anxiety symptoms; weaker connectivity was associated with lower cognitive ability. Panel (C) shows the significant association between infant attentional/regulatory ability and whole-brain theta connectivity; better attention/regulation was associated with stronger connectivity across all electrodes. Note that the dwPLI can take small negative values due to the debiasing procedure.



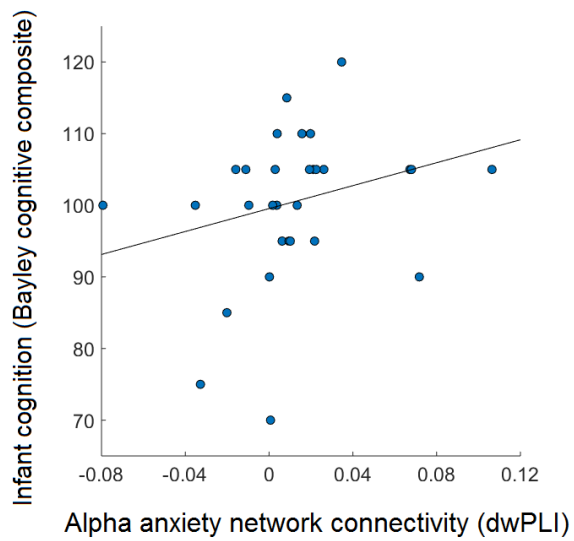




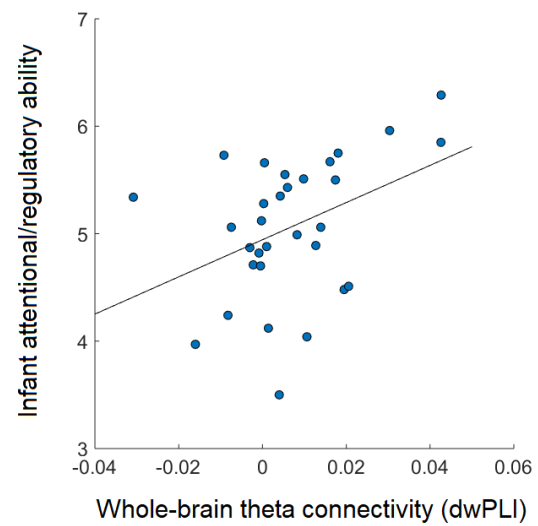
(A)



(B)



(C)



Effects of Maternal Psychopathology and Education Level on Neurocognitive Development in Infants of Adolescent Mothers Living in Poverty in Brazil

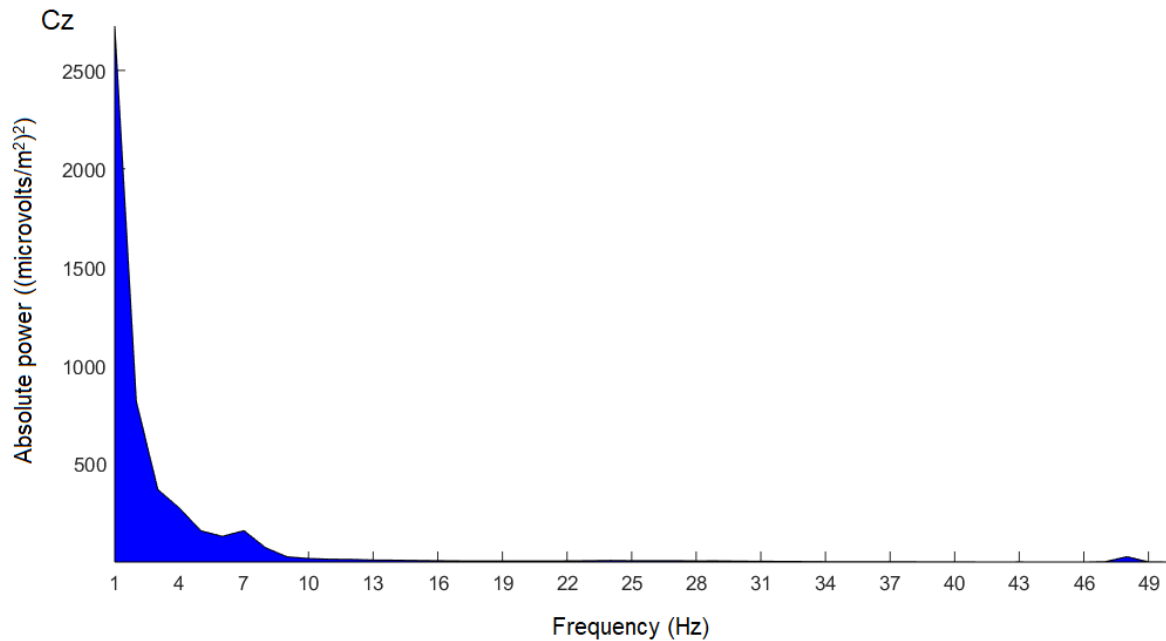
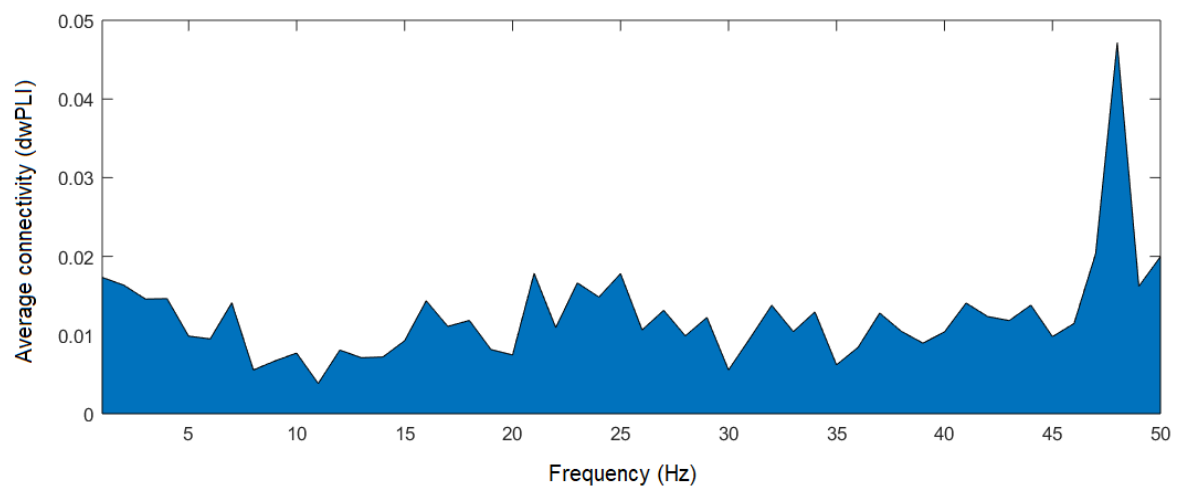
Supplemental Information

Retention analysis

Of the 50 infants taking part in this study, 31 (62%) provided usable EEG data and were included in analysis. Possible retention biases were investigated by comparing maternal and infant characteristics between the infants who did ($n = 31$) and did not ($n = 19$) provide usable EEG data. Maternal characteristics included symptoms of depression and anxiety assessed with the *Beck Depression and Anxiety Inventories (BAI & BDI; 1)*, ADHD symptoms measured by the *Adult ADHD Self-Report Scale (ASRS; 2)*, maternal level of education (classified in 5 categories ranging from *L1 – did not complete elementary school* to *L5 – completed university*), socio-economic status (classified as C/D or E according to the *Criteria for Economic Classification*, assessed by the *Brazilian Association of Research Companies questionnaire; 3*) and age (years) at recruitment. Infant characteristics (at age 6 months) included cognitive, language and motor ability assessed by the Cognitive, Language and Motor Composite scores on the *Bayley Scales of Infant Development-Third Edition (4)*, level of negative emotions and attentional/regulatory ability measured with the Negative Affect and Orienting/Regulation factor scores of the *Infant Behavior Questionnaire – Revised (IBQ-R; 5)*, sex, and age (weeks) at the 6-month assessment. A MANOVA comparing continuous variables between the EEG and no-EEG groups revealed no significant differences in maternal psychopathology, maternal/infant age, or infant cognition and behaviour (all $F \leq 3.13$, all $p \geq .08$, all $\eta p^2 \leq .064$). Similarly, Chi-square tests comparing categorical variables between the EEG and no-EEG groups revealed no significant differences in maternal socio-economic status, education level, or infant sex (all $\chi^2 \leq .637$, all $p \geq .31$).

Frequency band selection

We selected the theta (4-6Hz), alpha (6-9Hz) and gamma (30-50Hz) frequency bands for oscillatory power and connectivity analysis because these bands have been most robustly linked with cognitive functions and with effects of maternal psychopathology, adverse early environments and developmental problems in infants (see main text Background). To confirm our selection of these frequency bands was appropriate for our sample, we evaluated whether oscillatory power showed the typical frequency characteristics of infants of this age (i.e. a peak in power in the lower alpha range) and whether connectivity was particularly strong in a frequency band that we did not include in analysis. As Figure S1 shows, grand averaged power across all participants showed the typical alpha peak at ~7Hz, supporting our use of the predefined frequency bands in our sample. Figure S2 shows that the strength of connectivity (averaged across all participants and electrodes) varied little across the 1-50Hz frequency range, although we did see a peak in connectivity between 46-48Hz. Thus, our analysis focusing on the theta, alpha and gamma bands either include the frequency range containing the strongest connectivity (gamma band) or do not differ appreciably from the strength of connectivity in bands we did not examine (theta and alpha vs delta and beta). Since it is not recommended to calculate and analyse the dwPLI measure across narrow frequency bands because it can be unreliable (see e.g. 6) we did not conduct a supplementary analysis of the narrow-band 46-48Hz connectivity.

Figure S1 Average absolute power across participants plotted by 1Hz frequency step at site Cz**Figure S2** Average connectivity (dwPLI) across all electrodes and participants plotted by 1Hz frequency step

The debiased weighted phase lag index (dwPLI)

The debiased weighted phase lag index (dwPLI; 7) is an extension to a widely used measure of phase synchronisation, the phase lag index (PLI; 8). The PLI and dwPLI quantify phase synchronisation in terms of the extent to which two signals are coupled as reflected by the consistency of their (non-zero) phase lags (see (7,8) for further information). Both

measures range between 0 and 1, with values of 0 indicating the absence of phase synchronisation between the signals and values of 1 indicating perfect synchronisation between the signals. Extensions to the PLI introduced with the dwPLI include additional procedures to control for volume conduction and a debiasing procedure to correct for positive bias introduced by small sample sizes; as a consequence of the debiasing process, dwPLI values can sometimes take small negative values (see Figures 12-13 and description of the debiased weighting procedure on pages 1556-1557 of (7)). The negative-to-positive scale of the dwPLI can be interpreted in the same way as the traditional 0 to 1 scale, with smaller dwPLI values (e.g. those in the negative range) indicating weaker phase synchronisation and higher values (e.g. those above 0) reflecting stronger phase synchronisation.

Cluster-based permutation testing

Cluster-based permutation testing was used to identify clusters of electrodes at which oscillatory power in the theta, alpha and gamma frequencies was significantly positively or negatively associated with maternal psychopathology and education (first stage of analysis) and infant cognitive-behavioural measures (second stage of analysis). For each association of interest (e.g. theta power and maternal anxiety), the association between power at each electrode and the variable of interest was calculated to obtain a set of regression coefficients T . A threshold corresponding to $p < .05$ was applied to this set of coefficients to identify suprathreshold associations, which were then clustered into connected sets based on spatial adjacency. Cluster-level statistics from those sets were obtained by summing the T -values in each cluster and the resulting maximum cluster statistic was subjected to Monte-Carlo non-parametric permutation testing (using 1000 permutations) to obtain a p -value indicating significance level. This analysis method controls for multiple comparisons (i.e. the 80 power values from each of the 80 electrodes in power spectra) by reducing the data into spatially-

related clusters of electrodes, which are then subjected to statistical analysis (for further details see the tutorial paper presenting this method by (9)). In the main text, for each significant cluster-based permutation test, we report the maximum cluster statistic and its permutation p -value as well as an estimate of effect size; effect size (Cohen's d) was calculated by dividing the average cluster statistic by the square root of the sample, as suggested by Zalesky (personal communication, 2019).

Network based statistic (NBS)

NBS was used to identify oscillatory brain networks that were significantly positively or negatively associated with maternal psychopathology and education (hypothesis 1) and infant cognitive-behavioural variables (exploratory analysis). NBS first computes the association between each connection in adjacency matrices and the variable of interest resulting in a set of test statistics T , applies a primary threshold (T -value) to isolate connections with suprathreshold values, identifies topologically connected components (brain networks) among suprathreshold connections, and finally ascribes a p -value to identified networks via permutation testing. The choice of primary threshold T influences the likelihood of finding significant networks and the size (number of connections) in those networks. In the current study, we tested a range of primary thresholds: $T = 2.0$ (corresponding to $p \leq .05$), $T = 2.5$ (equivalent to $p < .018$), $T = 3.0$ (equivalent to $p < .005$), $T = 3.5$ (equivalent to $p < .002$) and $T = 3.6$ (corresponding to $p < .001$). Only networks that were significant at all primary thresholds were considered reliable and are reported in the main text Results; for those 'reliable' networks, we report the network connections and p -values for the primary threshold of $T = 3.0$ ($p < .005$) since this threshold yielded networks with neither extensive nor very few connections while providing strong statistical control. Preliminary analysis at the lowest primary threshold revealed dense, over-connected brain networks (≥ 78 nodes, ≥ 120 edges)

associated with variables of interest which may have reflected spurious connectivity; thus, an additional thresholding step was conducted in which all connections with non-significant (i.e. not significantly different from zero) connectivity values were excluded (set to zero) from adjacency matrices prior to NBS, as suggested in Zalesky *et al.* (10). In the main text, for each significant network identified with NBS, we report the average network statistic (T-value) for the significant connections in each network, the permutation p-value for the network as well as an estimate of effect size; effect size (Cohen's *d*) was calculated by dividing the average network statistic by the square root of the sample, as suggested by Zalesky (personal communication, 2019).

Intervention effects

All adolescent mothers in this study were taking part in a randomised controlled trial (RCT, trial number NCT0280718) of a nurse home visitation intervention designed to improve mothers' parenting skills and address their health needs and improve infant development (11). Of the 50 adolescent mothers included in the current analyses, 25 received the intervention and 25 care-as-usual. A range of assessments of maternal mental health and infant development were conducted at regular intervals (recruitment and 3, 6, 12, 18 and 24 months post-partum). Analyses examining the effects of the intervention on both maternal mental health and infant neurocognitive development over this two-year period are currently in preparation for publication (Fatori *et al.*, *In Preparation*). In the current study, our focus was on understanding how baseline levels of maternal psychopathology and education level affect infants' oscillatory brain dynamics in the first months of life, an issue that has not been studied previously in infants of adolescent mothers in developing countries. For these reasons, we have not included effects of the intervention on infants' oscillatory power and connectivity in the main analyses

of our current paper. However, since readers may be interested to know whether the intervention influenced infants' oscillatory activity, we summarise these analyses here.

Effects of the intervention were investigated by comparing oscillatory power (relative and absolute) and connectivity (connectivity in networks and whole-brain connectivity) in the theta, alpha and gamma bands between the Intervention ($n = 17$) and Control ($n = 14$) groups. Cluster-based permutation testing was used to compare oscillatory power between groups; network based statistic (NBS) was used to assess the presence of networks in which connectivity differed between groups. For oscillatory power, there were no clusters in which absolute or relative power differed significantly between groups (all $p \geq .99$). Likewise, for oscillatory connectivity, there were no networks in which connectivity differed significantly between groups (all $p \geq .20$).

While there were no effects of the intervention on oscillatory activity, it is possible that the intervention may have affected associations between maternal psychopathology and education and infants' brain development. Therefore, all analyses reported in the main text were conducted while covarying Group. In addition, we examined whether intervention Group interacted with maternal variables in associations with EEG activity. Specifically, we computed interaction terms between each maternal variable and Group (e.g. maternal anxiety $\times$ Group) and assessed whether this interaction term was significantly associated with infants' oscillatory power and connectivity. As reported in the main text, associations between maternal psychopathology and education and infants' power and connectivity were significant while controlling for Group and none of the interactions with Group were significant.

Supplemental References

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